Siddardth Pathipaka

Champaign, IL | Open to relocation, US-wide
US Work Authorization: STEM OPT (3 years, no sponsorship required through Feb 2029)
siddardth1524@gmail.com | +1(217) 255-0104 | GitHub | LinkedIn | Portfolio

SUMMARY

Aerospace quality engineer focused on composite structures, MRB workflows, and process data. M.S. Aerospace, UIUC (Dec 2025), with hands-on MRB and inspection experience at Tata Boeing on GE/Boeing flight-critical components. **Six Sigma Green Belt (CSSC)**. Built and deployed two **AIAG-compliant quality tools** (pFMEA, SPC) — both actively maintained on GitHub. Seeking Quality / MRB / Supplier Quality Engineer roles in commercial aerospace, space, and eVTOL.

SKILLS

Certifications: Six Sigma Green Belt — Council for Six Sigma Certification (CSSC) | Inspection & Quality Control in Manufacturing — NPTEL, IIT Roorkee

Quality / Analysis (Aerospace): AS9100D · AS9102 Rev B (First Article Inspection / FAIR) · NADCAP-aligned · pFMEA · SPC (Xbar-R, I-MR, p, u control charts; Cp/Cpk/Pp/Ppk indices) · 8D · RCCA · MRB Disposition · CAPA (AS9100-anchored) · NCR · CAR · NDT/NDI awareness (visual, tap test, ultrasonic) · Engineering Change / ECO · FAA Part 21 awareness · Quality Management System (QMS)

Composites & Manufacturing: Prepreg Layup · Autoclave Processing · Vacuum Bagging · Out-of-Autoclave Cure Methods · Cure Cycle Development · CFRP · Classical Lamination Theory · Hexcel/Toray prepreg system familiarity · GD&T (ASME Y14.5) · CMM Inspection · First Article Inspection · Lean Principles

Equipment & Tooling: CNC Machining · EDM · Tooling Qualification · Composite Layup Tooling · Autoclave Fixturing · Build Sequencing · SolidWorks · CATIA · AutoCAD

Tools & Software: Python · MATLAB · ABAQUS · CalculiX (FEA) · Streamlit · Git · Minitab

EDUCATION

Master of Science, Aerospace Engineering | University of Illinois Urbana-Champaign **Jan 2024 | Dec 2025**

Bachelor of Technology, Mechanical Engineering | VNR Vignana Jyothi Institute of Engineering and Technology **Aug 2019 | Apr 2023**

Relevant Coursework: *Advanced Composites Manufacturing · Mechanics of Composites · Space Structures · Finite Element Methods · Manufacturing Processes*

EXPERIENCE

Quality Assurance & MRB Intern — Aerospace Manufacturing (GE & Boeing Programs) | Tata Advanced Systems Limited **Oct 2022 | Mar 2023**

- Reviewed **CMM inspection reports against AS9102-style first-article-inspection requirements** on **450+ flight-critical components** for GE Aerospace and Boeing programs, validating GD&T callouts (position, profile, true position) against engineering drawings and flagging anomalies for MRB review.
- Contributed to **root-cause investigations on GE engine component nonconformances** — authored 5 Whys analyses and drafted sections of 8D reports under the CAR/CAPA lead, supporting NCR disposition closure and reducing NCR cycle time by ∼**22%** on the sample period.
- Built **Xbar-R control charts** on a CNC-machined positional feature; out-of-control signals at the original tool-change interval drove an RCCA recommendation to shorten the interval. After process-team implementation, defect rate on the feature dropped from **15% to under 3%** over the next production window.
- Built an **FMEA-based justification** for a supplier nonconformance, enabling an engineering-approved **use-as-is disposition** that prevented ∼**$3,000** in scrap and a 4-week supply delay.

Graduate Research Assistant — AE597 Composite Cure Process Development (Frontal Polymerization) | University of Illinois Urbana-Champaign **Aug 2024 | Dec 2024**

- **Validated a computational cure model** against experimental thermocouple data on frontal-polymerized thermoset composite specimens, achieving $<$**10% error on front velocity** and $\leq$**3°C on peak exotherm temperature** — qualifying the model as the lab's standard tool for process-parameter screening.
- Developed and validated a rapid **out-of-autoclave cure method** (frontal polymerization), reducing composite processing cycle from **8+ hours to under 5 minutes** in a research setting; enabling scalable OoA manufacturing exploration for aerospace composites.
- Enhanced solver stability by implementing **Olson heavy-set smoothing** and custom kernels, reducing curvature-induced oscillations by **40%** and enabling **2**$\times$ **mesh coarsening** without accuracy loss.
- Designed a **boundary-condition sensitivity framework** and conducted a stability study, improving model reliability and integrating results into the lab's pre-polymerization assessment workflow.

Manufacturing Lead, Brake & Suspension Systems | Team Solarians — Electric Solar Vehicle Championship (ESVC) **Aug 2019 | Mar 2023**

- Led brake and suspension manufacturing for a **30-engineer team** across a **3.5-year build cycle** — delivered **zero mechanical failures** across all dynamic events; scored **180.43/200 on the Cross Pad Test**.
- Designed and fabricated a **hydraulic disc brake system** and **double-wishbone independent front suspension**; owned the full build pipeline from raw material through vehicle integration, applying tolerance-stack discipline and first-article inspection at each stage.

- Achieved **7th place nationally** (608.8 points) through manufacturing-quality and reliability-focused build processes — outcomes attributed by the team to disciplined build sequencing and pre-event inspection protocols.

Manufacturing Intern | EQIC Dies & Moulds Engineers Pvt. Ltd. **Jun 2022 | Aug 2022**

- Verified **precision tooling tolerances to ±0.02 mm (GD&T)** and validated tool/component contact geometry (>**80%**) on production fixtures via **first-article inspection** — applying methods directly transferable to aerospace machining and composite layup tooling release.
- Mapped a **12-stage precision manufacturing workflow** (CNC, EDM, heat treatment, assembly), isolating inter-stage handoff points as the dominant source of dimensional tolerance stack-up.
- Conducted **CMM-based first-article inspection** on machined inserts, generating dimensional compliance reports against engineering drawings to authorize release for downstream finishing operations.

PROJECTS

SPC Manufacturing Quality Dashboard | Python, Streamlit | *Deployed* | GitHub **2024 | Present**

- Multi-page Streamlit app making classical SPC behavior visible across five aerospace manufacturing process streams, enabling detection of special-cause variation and process-capability quantification. Implements **Xbar-R/S, I-MR, p, and u control charts** with **Western Electric and Nelson rule detection**, Cp/Cpk/Pp/Ppk capability indices, and a real-time disturbance simulator. Validated with **83 unit tests per AIAG SPC 4th Edition**. Actively maintained.

FMEA Risk Prioritization Tool | Python, Streamlit | *Deployed* | GitHub **2024 | Present**

- Deployed web app digitizing the pFMEA process — scoring failure modes by severity, occurrence, and detection and exporting prioritized risk reports for design reviews and supplier audits. Built per **AIAG FMEA-4** with RPN scoring (S×O×D), Pareto 80/20 risk ranking, criticality flagging, and PDF/Excel export. Validated with **61 unit tests** on an aerospace CFRP composite panel dataset.

Virtual Composite Laminate Design & Optimization | Python, CalculiX FEA | GitHub **2024**

- Computational tool predicting the stiffness matrix and deflection response of fiber-reinforced composite laminates, enabling rapid ply-angle and stacking-sequence trade-off studies without physical prototypes. Implements a **Classical Lamination Theory engine with Simulated Annealing optimizer**, validated against CalculiX FEA achieving <**1% deflection error** and <**3% stress error** on an IM7/8552 CFRP plate.

SAMPE Composite Fuselage Competition — Manufacturing Lead, UIUC **Jan 2025 | May 2025**

- Fabricated a 24-inch composite fuselage via prepreg layup and autoclave cure (275°F, 40 psi), sustaining **2,700 lbf** at structural test. Applied pFMEA and standardized a 20 psi hold test protocol; achieved **sub-2% void content** on first article and **38% deflection reduction** via optimization.

SAEINDIA Aero Design Challenge — Team Lead **Mar 2022 | Oct 2022**

- Conducted CFD analysis using **ANSYS Fluent** on the SELIG 1223 airfoil, achieving **85% correlation** with theoretical predictions. Developed the manufacturing process plan for a 92-inch tapered balsa wood wing structure (layup sequence, tooling, build documentation). Project completed design phase.

Structural & Mechanical Properties of Borosilicate Glass–Alumina–Nano Copper Oxide Composites | Paper **Aug 2022 | Apr 2023**

- Fabricated and characterized **24 BS–Al_2O_3–n-Cu_2O composite formulations** via powder metallurgy (compaction + sintering at 850°C), investigating reinforcement effects on microstructure and mechanical hardness. SEM/EDS revealed distinct phase morphologies — chip-like alumina and spherical copper oxide nanoparticles dispersed in the glass matrix; XRD confirmed no reactive compound formation. Optimal composition (90% BS + 10% Al_2O_3) achieved **698.66 HV**, a **24.49% hardness gain** over pure borosilicate per **ASTM E384**. Results published in *Journal of The Institution of Engineers (India): Series D*.

LEADERSHIP & ACHIEVEMENTS

Leadership Roles

- **Technical Head — ASME Student Chapter, VNRVJIET** *(Aug 2022 – Mar 2023)* — led technical planning for engineering events including CADathon; coordinated workshops on CAD, design fundamentals, and manufacturing for **100+ undergraduate students**.
- **Manufacturing Lead — UIUC SAMPE Composite Fuselage Team** *(2024 – 2025)* — led layup, autoclave cure execution, and structural-test protocol on a 24" CFRP airframe; standardized hold-test and pre-test inspection procedure.
- **Manufacturing Lead — Team Solarians, Electric Solar Vehicle Championship** *(2019 – 2023)* — see Experience section for details (3.5-year build cycle, 30-engineer team).

Achievements & Recognition

- **7th Place Nationally** (608.8 pts) — Electric Solar Vehicle Championship 2023.
- **Cross Pad Test 180.43 / 200** — Team Solarians dynamic event score.
- **2,700 lbf structural test load** — SAMPE Composite Fuselage, UIUC 2025.
- **Six Sigma Green Belt** — Council for Six Sigma Certification (CSSC).
- **Inspection & Quality Control in Manufacturing** — NPTEL, IIT Roorkee.
- **AIAG-aligned quality tooling** — designed and deployed two open-source quality engineering applications (SPC Dashboard, FMEA Tool) actively maintained since 2024.
- **Peer-reviewed publication** — co-author on *Structural & Mechanical Properties of Borosilicate Glass–Alumina–Nano Copper Oxide Composites*, *Journal of The Institution of Engineers (India): Series D* (2024). DOI: 10.1007/s40033-024-00846-x